

INDUSTRIAL INTERNSHIP REPORT

Retail Analytics and Sales Prediction: Supermart Grocery

1. Executive Summary

This report details the comprehensive analysis of a grocery sales dataset focused on the Tamil Nadu region. The project utilizes Python-based Exploratory Data Analysis (EDA) and a Linear Regression model for sales prediction. Data indicates a consistent year-over-year sales growth from 2015 to 2018, with "Eggs, Meat & Fish" identified as the leading product category.

2. Project Overview

Objective

To identify purchasing patterns and growth trends across various cities in Tamil Nadu and develop a predictive model to forecast sales performance based on historical data.

Domain

Data Analytics & Machine Learning (Retail Sector)

3. Methodology

Tools: Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn)

- **Data Cleaning:** Removal of missing values and duplicates to ensure high data integrity.
- **Feature Engineering:** Conversion of date formats and extraction of temporal features (Day, Month, Year).

- **Data Preprocessing:** Label Encoding for categorical variables like City and Category to prepare the data for regression modeling.

4. Key Insights & Analysis

- **Category Performance:** "Eggs, Meat & Fish" is the top contributor to revenue, accounting for approximately 15% of total sales.
- **Regional Impact:** The West region consistently shows the highest sales volume.
- **City Rankings:** Kanyakumari, Vellore, Bodi, Tirunelveli, and Perambalur emerge as the top 5 cities by sales.
- **Growth Trend:** Significant upward trajectory observed over four years, with 2018 representing 33.3% of total recorded sales.

5. Predictive Modeling

A Linear Regression model was trained to forecast sales. The model achieved an **R-squared (R^2) value of 0.82**, indicating that 82% of the variance in sales is explained by the selected features. The Mean Squared Error (MSE) was recorded at 1758.26.

6. Strategic Recommendations

- **Inventory:** Prioritize stock for "Eggs, Meat & Fish" and "Snacks" during peak periods.
- **Expansion:** Target high-growth cities in the West and East regions for new distribution hubs.
- **Profitability:** Maintain the current average profit margin of ~24.8% by optimizing discount levels in lower-performing categories.

7. Conclusion

The analysis successfully maps the retail landscape for Supermart Grocery, providing actionable insights into consumer behavior. The developed regression model offers a reliable baseline for data-driven forecasting and future strategic planning.

